Orbital Dragons to Hunt Cryptids: Utilizing Non-Euclidean Geometry to Solve Busy Beaver

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DeepMath 2026 Extended Abstract

Abstract

The busy beaver value $BB(n)$ is the longest finite runtime of any n-state Turing machine started on a blank tape. $BB(5)$ was settled in 2024 with a computer-checked proof. $BB(6)$ is blocked by a handful of machines whose tapes secretly run Collatz-style arithmetic; nobody can prove whether they stop. We study two of these machines. Our approach is empirical first: we log the machine's behavior as a sequence of events, embed the event data in logarithmic, simplex, and hyperbolic coordinates, and find that the dynamics are piecewise affine in a five-integer state. We then prove this description correct in the Lean proof assistant, down to the machine's individual steps and with a minimal axiom footprint. The result is a machine-checked theorem: if a simple five-integer recurrence never enters an explicit finite set of bad configurations, the machine never halts. We also prove the converse pressure: entering a bad configuration forces the machine to halt. The open problem is thereby moved, without loss, from a Turing machine to a short arithmetic recurrence. We then derive the recurrence's renewal map in closed form, show its fatal set is a width-five window around the diagonal, and close the two mechanical proof routes with computational evidence: no congruence invariant up to modulus 384 excludes the window, and the branch dynamics has exponential factor complexity, so it is not conjugate to a rotation. Measuring the recurrence over its whole seed space shows the set of dying seeds has natural density zero at the Borel–Cantelli rate ($\approx 12/X$), so our machines' seeds are typical survivors. Surveying the entire holdout set with a two-ended harness (forward orbit against backward halt cone, regimes fitted in between) classifies all 1,094 open six-state machines: zero halts, zero alignments; four fixed-dimension machines, exactly the certified ones; an eleven-machine *array class* whose state spaces grow in dimension, for whose leading member we construct an exact word-rewrite calculus (a two-phase odometer with a block-trisection boundary rule) validated against 2×10^8 raw steps; and 307 single-law machines among 346 with regime structure. We frame the program on a verification spectrum whose proven endpoint is $\mathsf{MIP}^* = \mathsf{RE}$: kernel checking at the bottom, certified predictive emulation (this paper) in the middle, and entangled-prover verification of halting-type claims as the goal. Every empirical claim is validated against 10^9 raw machine steps, and the recurrence is tested for 10^8 iterations.

1 The problem, from zero

A Turing machine here is a finite table: six control states, a tape of 0s and 1s, and for each (state, read symbol) pair an instruction (write a bit, move left or right, change state) or nothing. If the table has no instruction for the current situation, the machine halts. $BB(n)$ asks: among all n-state tables that eventually halt from the all-zero tape, what is the longest runtime? The function is noncomputable [4, 3], so each new value is won machine by machine: every table either halts (run it) or must be *proven* to run forever.

The bbchallenge collaboration settled $BB(5) = 47{,}176{,}870$ with a proof checked in the Rocq proof assistant, "the first determination of a new Busy Beaver value in over 40 years and the first Busy Beaver value ever to be formally verified" [1, 2]. At six states a residual set of machines defeats all known automated deciders. The community calls them cryptids. The reason they are hard is concrete: their tape dynamics implement iterated arithmetic maps of the same character as the Collatz ($3x+1$) map: deterministic integer maps in which parity decides the next move, and where the question is whether one specific orbit ever hits a small bad set. The strongest known result for such maps is Tao's theorem that "$\mathrm{Col}_{\min}(N) \le f(N)$ for almost all N," for any function f diverging to infinity [5]. That theorem is about typical orbits. Busy beaver questions need the worst case: one orbit, followed forever. No general tool exists for that, which is why each cryptid is studied on its own terms.

We work on two six-state cryptids, a mirror-image pair we call dossier-2 and dossier-3. Machine one, written `1RB1RC_1LC1LC_1RE0LD_1LB1RF_---1RA_0RD1RD` in the standard notation, is the following table. Each entry gives (bit written, head move, next state); the machine starts in state A on an all-zero tape.

state	reads 0	reads 1
A	1, R, B	1, R, C
B	1, L, C	1, L, C
C	1, R, E	0, L, D
D	1, L, B	1, R, F
E	**(none: halt)**	1, R, A
F	0, R, D	1, R, D

The single empty entry is the only way to halt: the machine must be in state E reading a 0. Working backward through the table, state E is only entered from state C reading a 0, so any halt must pass through the configuration "C reads 0 with another 0 to its right." We call this the *probe event*. Everything that follows is one long argument that the probe event never happens.

2 What we do

Our method has three stages. Watch, compress, certify.

Watch. We run the machine and record only the moments that matter. For these machines there is a natural clock: a recurring configuration we call a spawn entry. Between two spawn entries the machine performs one bounded "episode" of work. Sampling at spawn entries turns a raw trajectory of 10^9 steps into a clean sequence of a few million events.

Compress. At every spawn entry the tape has the same form: a solid block of 1s with exactly two 0-cells ("holes") inside it, the head at the block's right end. A concrete instant of the certified orbit (machine at position r, dots are 0s):

$$\cdots 0\,0\,\underbrace{1\,1\,1\,1\,1\,1}_{a}\,\underset{h_1}{0}\,\underbrace{1\,1\,1\,1}_{b-1}\,\underset{h_2}{0}\,\underbrace{1\,1\,1\,1\,1\,1}_{c-1}\,\underset{r}{\downarrow}\,0\,0\cdots$$

So the whole configuration is five integers: the block's endpoints l, r, the hole positions h_1, h_2, and a tag; we write $a = h_1 - l$, $b = h_2 - h_1$, $c = r - h_2$ for the gaps. Between spawn entries the machine performs one of seven "episodes," each later certified with its exact step count:

episode	steps	effect on the tape
spawn	7	extend block by 1, dig a fresh hole at $r-2$
crawl	10 per cycle	the walking gait; moves 2 cells, carries a hole
cross	13	absorb a hole, re-dig it 4 cells right
terminate	13	fill a hole, plant it 6 right (the probe events)
fuse	19	merge adjacent holes, plant 9 right
edge	10 or 13	extend the left end by 1 or 3 (renewal)
sweep	2 per cell	return home; holes cross for free

For instance, the crawl is a fixed ten-step loop: the machine marches left three cells twice (writing $1, 1, 0$ each time), then backs up four; the net effect is two cells of travel and a 0 that the walker carries along three cells behind itself. The danger state C reads a 0 *only* inside terminate and fuse episodes, and there the theorems show the cell to its right always holds a 1, so the probe event cannot fire. Plotted in raw coordinates, the event data looks like growth curves with no visible law (Figure 1). Plotted in the right coordinates the same data is piecewise linear: componentwise $\log(1+x)$ straightens each regime, the projective simplex $(a, b, c)/(a+b+c)$ removes scale, and hyperbolic and spherical renderings of those two charts expose the orbit's nested renewal structure (Figure 2). Straight segments are one regime; corners are regime changes.

These coordinate choices are quantum-inspired data science in a precise, bounded sense, and each has published lineage. The log chart is a dequantization: in Viro's words, "on logarithmic paper some real algebraic curves look like smoothed broken lines" [8], and Litvinov states the limit exactly: "tropical mathematics can be treated as a result of the so-called Maslov dequantization of the traditional mathematics over numerical fields as the Planck constant $\hbar$ tends to zero taking imaginary values" [9]. Our piecewise-linear regimes are that classical limit, observed in a Turing machine's arithmetic. The simplex chart with its Fisher–Rao metric is the classical face of quantum state geometry, made literal by the amplitude map $p \mapsto \sqrt{p}$ onto the positive octant of the sphere. The hyperbolic chart follows the machine-learning use of negative curvature for hierarchies, which "allows us to learn parsimonious representations of symbolic data by simultaneously capturing hierarchy and similarity" [10]; our nested renewal generations are exactly such a hierarchy. Two further disciplines carried over from quantum simulation practice: our vector encodings of Collatz-like steps are *reversible* maps (the circuit-model requirement that every operation be invertible), which is what makes backward orbit search, the classical analogue of uncomputation, available for the halt cones of the census below; and candidate states are evolved in ensembles. We state the boundary plainly: no quantum computation is performed anywhere in this pipeline, and no quantum advantage is claimed; "quantum-inspired" names the origin of the geometries and the disciplines, not a resource. The underlying view of a Turing machine as a dynamical system goes back to Moore, for whose generalized shifts "many simple questions about their dynamics are undecidable" [6], and to Kůrka [7].

Certify. The picture suggests a law; the law must then be proven about the actual machine, or it is only a picture. We first validate it empirically and fail-closed: a predictor that knows the conjectured law must reproduce every event, with exact positions and exact tape, over 10^9 raw steps per machine, and any miss aborts. It reproduces all of them. We then prove the law in Lean 4, using only the kernel's minimal axioms (`propext`, `Quot.sound`; no classical choice). The certified account consists of: seven theorems, one per episode type, each giving the episode's exact step count, exact output shape, and safety; two inductions that handle the episodes' unbounded middle parts; a concrete certificate for the machine's startup (235 steps for one machine, 2,044 for the other); and a transport theorem showing that a five-integer function `metaStep`, iterated from the startup state, predicts the machine forever.

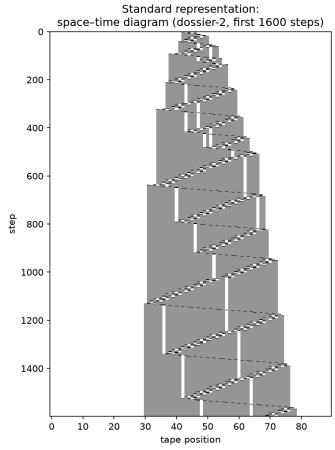


Figure 1: Raw view: the machine's first 1,600 steps (dark = 1s, one row per step). Nothing about the long-run law is visible here.

3 The result

Halting, for these machines, can only happen one way: a specific control state must read a 0 with another 0 beside it. The certified episode theorems show this situation never arises while the five-integer iteration stays inside an explicit guarded region. Writing SafeMetaOrbit for the statement that the iteration never leaves that region:

Theorem 1. *For each machine, checked by the Lean kernel at axioms* {propext, Quot.sound}*:* SafeMetaOrbit $\to \neg$RawHalts(blankStart)*.*

Two facts make this reduction meaningful rather than cosmetic. First, it is complete: startup, one exceptional episode, and every future step of the machine are covered by theorems, so nothing about the machine remains to be understood except the arithmetic. Second, it is sharp: we constructed the configurations just outside the guarded region and ran the machine from them; it halts within about fifty steps. Leaving the region is not merely unproven-safe. It is fatal. So SafeMetaOrbit *is* the halting problem, restated as arithmetic.

The arithmetic is explicit. Inside the main regime the iteration is $(a, b, c) \mapsto (a{+}6,\ b{-}2,\ c{-}2)$, so $\gamma = c - b$ is conserved while the counters drain. When b or c runs out, a renewal move re-seeds the triple by subtracting and swapping counters, with parity choosing the branch: part Euclidean algorithm, part Collatz. Figure 3 shows the renewal process as its own dynamical system: the return map of γ, with the fatal values $\gamma \in \{1, 3, 5\}$ marked. The recurrence avoids the band for 10^8 iterations (about 10^{15} machine steps). Proving it does so forever is the open problem, stated entirely in elementary arithmetic.

We characterized that problem further. Sampling the orbit at its renewal section reduces the whole dynamics to a two-counter map, verified exactly against the certified orbit: with $(\alpha, \beta) = (6, a{-}1)$ or $(9, a{-}2)$ by the parity of a,

$$R(a,b) = \begin{cases} (a+6,\ b-3) & b \text{ even}, \\ (\alpha + 3(b-3),\ \beta - (b-3)) & b \text{ odd},\ b < \beta, \\ (\alpha + 3\beta + 3,\ b - \beta - 4) & b \text{ odd},\ b > \beta. \end{cases}$$

The case $b = \beta$ never occurs on the verified orbit, against which the map is checked transition by transition. Its fatal set is geometrically simple: death occurs exactly when the two counters arrive

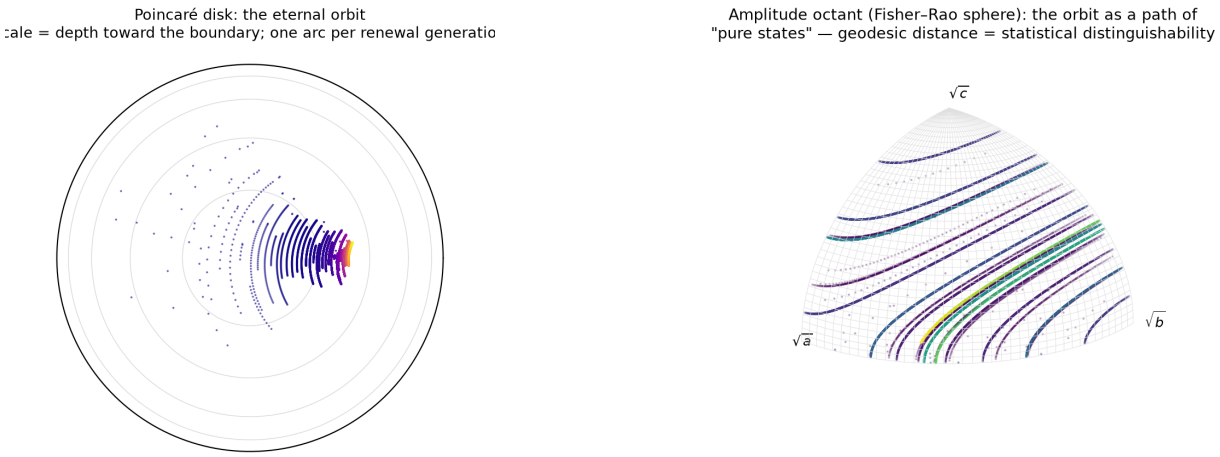


Figure 2: The same machine, 100,050 events, in two non-Euclidean coordinate systems. Left: the Poincaré disk. Shape runs along the boundary direction, log-scale is hyperbolic depth, and each renewal generation of the orbit sits one nested arc deeper. Right: the amplitude octant. The normalized state $(a, b, c)/(a+b+c)$ mapped by $p \mapsto \sqrt{p}$ onto the Fisher–Rao sphere, so the orbit becomes a path whose geodesic length measures statistical distinguishability. Straight arcs are single regimes; corners are regime changes.

almost equal ($b - \beta \in \{1, 3, 5\}$, plus two boundary cases). Two mechanical proof routes are then closed by computation. A sound residue-abstraction search finds a genuine mod-3 conservation law but shows that no congruence invariant up to modulus 384 excludes the fatal window. And the branch-symbol sequence has factor complexity growing like 1.6^n, where a rotation coding would give $n + 1$, so the dynamics is not conjugate to a rigid system, and continued-fraction rigidity arguments do not apply as stated. Over 10^5 renewals the orbit's minimum distance above the fatal window is 4: it grazes death and survives, so no proof can rely on a structural buffer. What remains is a pointwise avoidance statement for a pseudorandom multiplicative map, the Collatz situation, with three mitigations: the map is explicit and total, the fatal set is a finite union of lines near the diagonal, and the certified reduction stands under any future argument. The natural next language is hyperbolic: renewal steps are integer 2×2 matrix moves, and a near-tie of the counters is a cusp excursion on the modular surface, so the conjecture asks that a specific geodesic avoid three explicit horoballs forever [11].

Almost all seeds survive. We also measured the map R over its whole seed space, not just our orbit. Enumerating every seed in $[X, 2X)^2$ exactly for $X \le 256$, and sampling to $X = 2^{16}$, the fraction of seeds whose orbit ever dies is $\approx 12/X$: the dying set has natural density zero, at the rate a Borel–Cantelli heuristic predicts. The mechanism is visible in the data. Every observed death occurs within the first fifty renewals; the offset $b - \beta$ at the diagonal is uniformly distributed, fatal values included; and sizes grow by a factor $e^{0.134}$ per renewal, so the per-renewal hazard $\approx 6/\text{size}$ sums geometrically: death is an early-game phenomenon, and survival compounds. Our machines' seeds are therefore *typical survivors*, not exceptions. (We record one measurement artifact. An early instrument logged offsets only on steps actually taken and reported zero fatal offsets in 2×10^5 events; this is survivorship bias in textbook form, since a fatal offset is a death and never generates a step. Corrected instrumentation restored exact agreement with the uniform prediction.) A rigorous almost-all theorem needs one missing piece: the tail-hazard bound along orbits, an invariant-measure argument in the style of Tao's Collatz result, with the advantage that

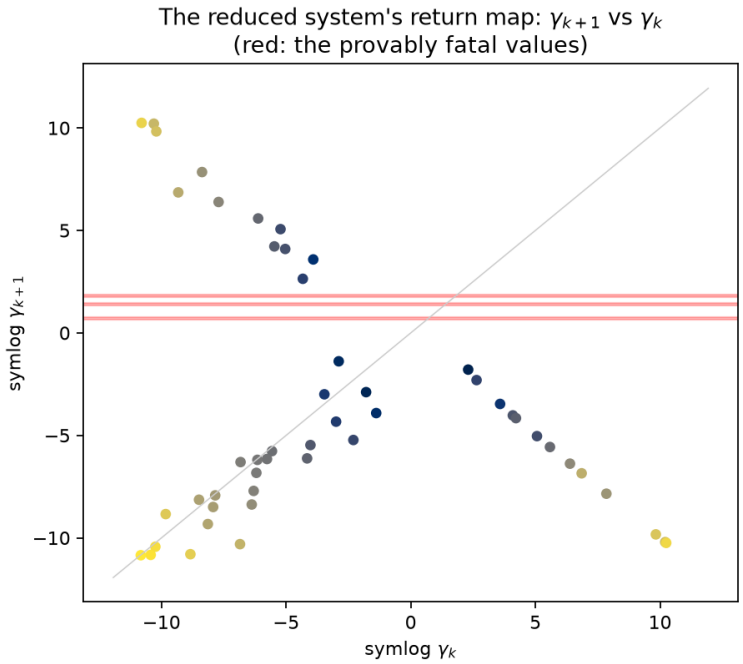


Figure 3: The reduced problem as a dynamical system: the return map of the renewal invariant, γ_{k+1} against γ_k (symlog axes), over the first fifty renewals. The red bands are the values proven fatal. The conjecture: the map's orbit never lands in a band.

R is explicit, total, piecewise affine, and its hazard is measured uniform.

4 The census: what the remaining 1,094 machines are

The pipeline above certifies individual machines. We also ran it, in survey form, across the entire BB(6) holdout set, and the result is a classification: to our knowledge, the first structural census of everything that remains open at six states.

The instrument is a three-part harness modeled on a two-prover frame (whose provenance the next section makes precise): a *start machine* (forward emulation from the blank tape), an *end machine* (the backward halt cone: every partial configuration that could precede a halt within k steps), and a *regime machine* that detects and fits the piecewise laws a run moves through, using second differences of extreme-event times. A machine halts if and only if its forward orbit ever aligns with its backward cone, so the verdict compiles from the three reports. Across all 1,094 holdouts: zero halts, and zero alignments; at every sampled instant, every holdout is at least 18 raw steps from any halting event. (For dossier-3 the two ends are provably incompatible in a stronger sense: all 134 backward families at depth 24 require three 0s within a six-cell window, a pattern no certified forward shape can produce; the statement is finite and checkable.) The census then splits the set:

- **4 machines** (Antihydra and our dossiers 1–3) reduce to *fixed-dimension* integer maps, a bounded tuple of counters with certified episode theorems. These are the machines the previous sections certify, and the sweep found no others: ranked by regime cleanliness, the top 25 candidates yield 11 machines with exact fail-closed orbit specifications, and *every one of*

the eleven fails both fixed-shape gates (bounded hole count, bounded run-length dimension). The certified four are, up to this ranking's reach, the only fixed-dimension machines in the cohort.

- **The array class.** Those 11 exact-but-unbounded machines are the new frontier: their tape shapes are growing *vectors* of blocks; the state space itself gains a dimension every few regimes. Their orbit specifications are exact but fragment into hundreds of rules of support one or two; the law is real but lives on a space of growing dimension.
- **307 machines** are governed by a single affine law in the regime harness (the structured third of the cohort, 346 with detectable regime structure); the rest show no structure at the probed horizons.

For the top array-class machine (`1RB1RD_1LC0LC_0RD0LB_1RF0LE_0RD---_0RA1RC`) we then built the array-class analogue of the interaction calculus: an exact *word-rewrite system*. Its tape at each canonical event is a word of 1-blocks of lengths 1, 2, 3 with single-0 gaps; a sweep of the head is a sequential transducer over that word, and the whole dynamics compresses to five interior rules (copy, absorb, cursor, insert, and one parametric regroup family with cost $8 + 6k$), two parametric boundary families, one of which is a literal *block trisection* ($3 \mapsto 1{,}1{,}1$), and two derived laws: a deposited 2-block acts as a *cursor*, restarting the next pass at its own position, and whether a pass ends in a restart is read off the output word. Semantically the machine is a two-phase odometer on $\{1, 3\}$-words: the 3s are carries, the cursor is the carry boundary. The system is validated the way the certified machines were before their Lean compilation. Rules learned on 4×10^7 raw steps reproduce 2,179 of 2,186 held-out segments over the next 1.6×10^8 steps *exactly* (output word, arrival type, step count, head position), with zero chaining violations; every exception is a flagged boundary-window event, never a silent misprediction, and re-learning over the full horizon leaves every decision key single-valued (zero conflicts across 3,956 segments). The sharpest check runs the machine itself on the calculus's own *predictions*: configurations forecast up to 139 sweeps beyond any simulated history, written onto a literal tape and handed to the raw machine, reproduce every forecast exactly, step counts included. This is predictive emulation (certify the law, then verify forecasts) standing, for the first time in this program, on a state space of growing dimension. The cohort boundary is sharp on both sides: a tenfold budget increase changes no verdict (all fourteen ranked candidates that failed the exact-spec gate still fail it, so the eleven-machine array class is complete under this criterion), and porting the formalism without machine-specific rules across the other ten machines places one at a 99.8% architecture match with 82.4% of held-out segments reproduced from literal rules alone, and two more at 99.3% and 98.6% exactness where extraction succeeds.

The classification also carries a quantitative signature. The abstract backward cone of dossier-3, the tree of renewal-level states dying within k steps, has level counts $9, 19, 31, 50, 81, \ldots$ satisfying $L(k) = L(k{-}1) + L(k{-}2)$ *exactly* from depth two on: a Fibonacci tree, branching at the golden ratio, and the same $\varphi \approx 1.618$ appears independently as the forward factor complexity. The mechanism is visible: a terminate step flips a parity, so no branch word contains two consecutive terminates, precisely the constraint that generates Fibonacci counting. The standing, falsifiable prediction for the array class: each machine's death tree satisfies a small integer recurrence with a Pisot-like constant determined by its episode-alternation rules, the golden ratio being the founding case. The rank-1 machine supplies the second data point. Its cycle grammar inverts in closed form: the number of one-sweep preimages of an interior word equals $\prod_i 2^{\max(0,\, t_i - 2)}$ over its $(1, 3)$-train lengths t_i, so this death tree branches 2-adically in train lengths where dossier-3's branches at φ.

5 The verification spectrum: where this sits, and the goal

We place this work on a spectrum of what "verifying a machine" can mean; both endpoints of the spectrum are now theorems.

Starting verification is the ground-up kind: replay and check every step. Our Lean development is this, industrialized: the kernel checks each of the 10^4 startup steps individually, and the trust surface is two axioms and a proof checker. It is complete and unconditional, but it can never outrun the machine: verifying n steps costs n.

Predictive emulation is the middle of the spectrum and the substance of this paper. Instead of replaying steps, certify the machine's *law*, the episode theorems and the five-integer map, and then verify *forecasts*: where the next probe event can occur, what the configuration will be 10^6 steps from now, whether halting is reachable from the current shape. One certified episode theorem covers unboundedly many raw steps; the certified recurrence covers all of them. Verification cost decouples from runtime. This is what lets two machines' entire futures be checked against one explicit arithmetic conjecture.

Ending verification is the horizon, and we take it as the program's goal: $\mathsf{MIP}^* = \mathsf{RE}$ [12]. A classical polynomial-time verifier, interacting with two entangled provers, can verify *any* recursively enumerable statement, with the provers rather than the verifier bearing the unbounded cost; in the authors' words, "there is an efficient reduction from the Halting Problem to the problem of deciding whether a two-player nonlocal game has entangled value 1 or at most 1/2" [12]. Non-halting statements such as ours live one level up (they are Π_1; zero-gap variants of MIP^* reach Π_2^0 [13]), so the entangled-prover frame in principle spans exactly the class of claims the busy beaver program generates. The alignment we draw is this: kernel verification anchors truth at the bottom; predictive emulation compresses it in the middle; and $\mathsf{MIP}^* = \mathsf{RE}$ says that at the top, verification of halting-type claims is possible *in principle* with entanglement as the resource. Our quantum-inspired toolkit (reversible vector maps, state-space geometry) borrows disciplines from that endpoint; the open engineering-of-mathematics question is how much of the middle rung, certified forecasts, can be pushed toward it, replacing replay with interaction. Nothing in our results depends on this framing; it states the destination the pipeline is pointed at.

6 Why this may interest this community

We offer two reasons. The first is the workflow. This project is an instance of the emerging human–AI–kernel pattern in mathematics, in which AI systems draft and compile proofs at scale, a proof kernel arbitrates, and humans own claims; this is the pattern of Tao's Equational Theories Project, which "successfully determined all 22 028 942 edges of the implication graph between the 4694 simplest equational laws on magmas," and of the new Palomar registry for Lean-verified results [14, 15]. All of our Lean was drafted by an AI assistant from validated empirical specifications; the kernel checked roughly fifteen thousand lines; the generators are convenience, not trust surface. The second reason is methodological kinship. Finding that a high-dimensional trajectory is governed by a low-dimensional piecewise-affine law, by choosing coordinates in which the structure is linear, is the daily business of representation learning. Here the same move is carried one step further: the discovered representation is not evaluated on held-out data but proven exactly correct, step by step, against the system that generated it.

Scope. We claim no unconditional non-halting result. The coordinate charts guide discovery and exposition; no theorem depends on them. Empirical horizons are what we ran, not what is true.

Acknowledgments. The Lean developments, empirical validation, and proof generators were built in collaboration with an AI assistant (Claude, Anthropic). Every proof is checked by the Lean 4 kernel at the stated axioms.

References

[1] The bbchallenge Collaboration. Determination of the fifth Busy Beaver value. arXiv:2509.12337. "Our result marks the first determination of a new Busy Beaver value in over 40 years and the first Busy Beaver value ever to be formally verified." The kernel-checked standard our Lean developments follow; source of the BB(6) holdout set our census classifies.

[2] The bbchallenge Collaboration. Turing machine deciders, part I. arXiv:2504.20563. "Each decider is specialised in recognising a particular type of non-halting behavior." The automated paradigm whose residue at six states is our 1,094-machine cohort.

[3] P. Michel. The Busy Beaver Competition: a historical survey. arXiv:0906.3749. "He used Turing machines to give explicit definitions for some functions that are not computable and grow faster than any computable function." The historical record, including early observations of Collatz-like champions.

[4] S. Aaronson. The Busy Beaver Frontier. *SIGACT News* 51(3), 2020. "The BB function makes the concepts of computability and uncomputability more vivid than anything else ever invented." The open-problem landscape this submission addresses.

[5] T. Tao. Almost all orbits of the Collatz map attain almost bounded values. arXiv:1909.03562. "One has $\mathrm{Col}_{\min}(N) \leq f(N)$ for almost all N" (for any f diverging to infinity). The strongest known result for the map class our machines implement; our almost-all-seeds measurement mirrors its shape, and the missing tail-hazard bound we identify is an invariant-measure argument of this kind.

[6] C. Moore. Generalized shifts: unpredictability and undecidability in dynamical systems. *Nonlinearity* 4 (1991). "Many simple questions about their dynamics are undecidable." The founding statement of the machine-as-dynamical-system view our coordinate charts operationalize.

[7] P. Kůrka. On topological dynamics of Turing machines. *Theoret. Comput. Sci.* 174 (1997). The topological-dynamics formalism (moving-head and moving-tape models) underlying the modern study of machine trajectories.

[8] O. Viro. Dequantization of real algebraic geometry on logarithmic paper. arXiv:math/0005163. "On logarithmic paper some real algebraic curves look like smoothed broken lines." Why the log chart straightens our regimes: the piecewise-linear law is the tropical limit of the machine's arithmetic.

[9] G. Litvinov. The Maslov dequantization, idempotent and tropical mathematics. arXiv:math/0507014. "Tropical mathematics can be treated as a result of the so-called Maslov dequantization of the traditional mathematics over numerical fields as the Planck constant $\hbar$ tends to zero taking imaginary values." Names the limit precisely; our quantum-inspired toolkit borrows a discipline, not a metaphor.

[10] M. Nickel and D. Kiela. Poincaré embeddings for learning hierarchical representations. NeurIPS 2017. "The underlying hyperbolic geometry…allows us to learn parsimonious representations of symbolic data by simultaneously capturing hierarchy and similarity." The machine-learning precedent for our hyperbolic chart; nested renewal generations are hierarchical data.

[11] C. Series. The modular surface and continued fractions. *J. London Math. Soc.* 31 (1985). The geodesic-coding dictionary behind our restatement of the five-integer conjecture as horoball avoidance.

[12] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen. $\mathsf{MIP}^* = \mathsf{RE}$. arXiv:2001.04383; *Comm. ACM* 64(11), 2021. "There is an efficient reduction from the Halting Problem to the problem of deciding whether a two-player nonlocal game has entangled value 1 or at most 1/2." The proven endpoint of our verification spectrum and the bridge motivating the two-prover census harness.

[13] H. Mousavi, S. Nezhadi, and H. Yuen. Nonlocal games, compression theorems, and the arithmetical hierarchy. STOC 2022; arXiv:2110.04651. "Deciding whether the quantum value of a two-player nonlocal game is exactly equal to 1 is complete for Π_2." Places non-halting claims within reach of the zero-gap entangled-prover frame.

[14] T. Tao et al. The Equational Theories Project. arXiv:2512.07087. "The project successfully determined all 22 028 942 edges of the implication graph between the 4694 simplest equational laws on magmas." The flagship instance of the human–AI–kernel workflow this project follows.

[15] Palomar: a registry of Lean-verified mathematics, 2026. `palomar-registry.org`. "Palomar records mathematical claims from fixed versions of their source files, checks their proofs with Lean, and publishes the exact statement, the libraries it uses, and the review's comments." The channel where our certified reductions are externally checkable at a pinned commit.